Akoizo Original Practice - Form 3

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Doubling wheel radius with the same axle and string does what to driven distance?

A. Doubles it

B. Halves it

C. Quadruples it

D. No change

Answer: _____

2. (2 points) What is the Distance Score for a Failed Run?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. 2500 points

B. 0 points

C. 150 points

D. 300 points

Answer: _____

3. (2 points) Why must the stopping setting change for a curved path?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. The target moves

B. Curves shrink wheels

C. The arc is longer than the straight-line distance

D. The cans slow the vehicle

Answer: _____

4. (2 points) Why recheck calibration on the competition floor?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. Wheels grow

B. The target changes the rules

C. Surface changes skid distance and wheel grip

D. Gravity changes

Answer: _____

5. (4 points) List three places energy is lost between the mass and the wheels.

Answer: _____
_____
_____

6. (4 points) Why build calibration tables for both straight and bonus runs?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

Answer: _____
_____
_____

7. (4 points) Why is a progressive slowdown better than a sudden stop?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

Answer: _____
_____
_____

8. (4 points) Why must wheel diameter be measured precisely?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

Answer: _____
_____
_____

9. (4 points) How is the final score determined from two runs?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

Answer: _____
_____
_____

10. (4 points) Why master straight runs before the bonus?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?

Answer: _____
_____
_____

12. (2 points) Which setup is better?

Supplied synthetic observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

A. 2.0 kg mass

B. No difference

C. 1.2 kg mass

Answer: _____

13. (2 points) Should the team try the Can Bonus on Run 2?

Supplied synthetic observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. No; Run 2 replaces Run 1

B. Yes; the expected score is lower and Run 1 is banked

C. No; the bonus never helps

Answer: _____

14. (2 points) Define “Backstop” in the context of Scrambler.

Answer: _____
_____
_____

15. (2 points) Define “Distance ratio” in the context of Scrambler.

Answer: _____
_____
_____

16. (2 points) Define “Vehicle Distance” in the context of Scrambler.

Answer: _____
_____
_____

17. (2 points) Define “Inside Can” in the context of Scrambler.

Answer: _____
_____
_____

18. (2 points) Define “Brake” in the context of Scrambler.

Answer: _____
_____
_____